

About

About the Biotechnology Program

Biotechnology is the transformative science of leveraging living systems - cells, organisms, and biological processes to pioneer innovations that enhance human health, nourish our planet, and power sustainable progress. Biotechnology continues to expand rapidly with new discoveries at a breath taking pace. Our Program in B. Tech Biotechnology equips students with the scientific and technological expertise to address pressing global challenges and develop technological solutions to problems in health care, nutrition, environment, agriculture, and energy. Our curriculum offers multidisciplinary approach where students will learn engineering subjects such Mechanical, Electrical, Electronics, Computer Science and their application in addressing biological issues which are the emerging areas of Biomechanics, Biosensors, Bioelectronics, Computer aided biology etc., to become bioengineers apart from Life Sciences subjects. Other cutting edge areas students will be exposed and specialize are in Medical Biotechnology & Bioinformatics, Plant Biotechnology, Animal & Environmental Biotechnology, Bioprocess Engineering and Nano-biotechnology and other emerging areas. Our pedagogy encourages imbibing practical skills that can be honed through project-based learning and by studying real-world case scenarios. Department nurture tomorrow's bioengineers and technologists who can design, test, and manufacture biological systems and bio-based products for diverse applications. This program, a merger of science, engineering and business, demands a multi-disciplinary workforce skilled in basic-research, product development, regulatory affairs and commercialization.

Nurturing Leadership: Cultivating critical thinkers, entrepreneurs, and global problem-solvers who can lead India's as well global biotech revolution.

Laboratory Infrastructure: Department has established Cell Biology & Microbiology Lab, Biochemistry Lab, Molecular Biology & Genetic Engineering Lab, Bioinformatics, Fluid Mechanics and Heat Transfer Labs. The labs which will be established shortly, are Bioprocess Engineering Lab, Plant Biotechnology lab, Immunology Lab, Downstream Processing Lab, and Animal Biotechnology Lab etc. The established labs have good infrastructure (equipment's) where students and faculty could conduct real time projects and carry R&D activities apart from curriculum bound hand on experience with experiments.

Memorandum of Understanding:

VNR VJIET has signed an MoU with Center for Cellular and Molecular Biology (CCMB), Habsiguda, Hyderabad for student related projects and R & D activities.

Future Prospects and Placements

Apart from teaching, at VNR VJIET we counsel and guide the students based on their potential which domain of biotechnology they can succeed as a part of career counselling.

Biotechnology

Biotechnology is one of the highly skilled field and Biotechnologists and Bioengineers could earn in par with any engineering professionals besides serving the nation.

Higher Studies: Students with B. Tech could target higher education and pursue M. Tech / MS / Integrated PhD in Indian & Foreign Universities or R&D organizations which in the long run has more growth potential.

Employment: In Academic / R&D / Government organizations:

Students who complete their B. Tech / M. Tech / MS / Ph.D in Biotechnology can find jobs in ? IIT?s, NIT?s, Central & State Universities etc (Academic); CCMB / CDFD / ICRISAT / IICT / DRDO / CSIR / ICAR / ICMR Laboratories etc (R&D), Pollution Control Board, Patent Office, Central & State run Forensic Labs (Government).

Industry Placements: Reputed Multinational Biotech industries

like Reliance Life Sciences, Novartis, Bharath Biotech, Biocon, Serum Institute, Dr.Reddy?s Labs, Clinical Research Organizations etc in India.

IT / Software Jobs: Bioinformatics based IT jobs in MNCs like

Infosys, TCS, Wipro, CTS etc apart from they are directly eligible for software jobs as they study major computer science subjects which are prerequisite.

Job opportunities in industries and research organizations

related to Agricultural & Plant Biotechnology, Medical & Pharmaceutical Biotechnology, Environmental Biotechnology, Bioinformatics, Food Science & Technology and in Chemical / Bioprocess Engineering etc

Industry ? Institute Collaboration for Student Projects

R&D Based: University of Hyderabad,

Osmania University, NALSAR University, JNTUH, Center for Cellular and Molecular Biology, Hyderabad, Centre for DNA Fingerprinting and Diagnostics (CDFD), National Institution of Nutrition, Indian Institute of Chemical Technology and Institute of Genetics.

Biotechnology

Industry: Well Known Biotechnology based

Industries like Novartis, Virchow Biotech, Bharat

Biotech, Vimta Labs, Indian Immunologicals, Sudharshan Biotech and Biocon
(Bangalore), Serum

Institute (Pune) etc for pursuit of student projects.

In India, Biotechnology industry has its presence more in cities like

Hyderabad, Bangalore,

Pune, Mumbai, Chennai, Kolkata, Delhi, Ahmadabad etc